

EduSync AI - Enterprise School & College Management System

Official User Manual, Operational Guide & GUI Walkthrough

Version 3.2.0

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1. Introduction & System Overview

EduSync AI is a next-generation Cloud ERP and Academic Operating System designed for K-12 schools, high school academies, vocational colleges, and university networks. Built with a high-performance **Liquid Glass** user interface, EduSync AI unifies all administrative, academic, financial, and facility management operations under a single platform.

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Key Differentiators

- **Dual-Engine Architecture:** High availability PostgreSQL backend with automatic in-memory fallback ensure uninterrupted operational uptime.
 - **Generative AI Integration:** Native integration with Google Gemini 1.5 Flash powers an interactive AI Student Tutor, automated academic at-risk predictions, and 1-click English-to-Arabic SOP translation.
 - **Bilingual & RTL Ready:** Instant dynamic switching between English and Arabic (`dir="rtl"`) for guidelines, forms, timelines, and interface components.
 - **Multi-Tenant & Multi-Role:** Standardized role-based access for 7 distinct campus personas across dedicated web portals.
 - **Native INR (₹) Financial Management:** Enterprise pricing tiers, fee invoice generation, payment transaction tracking, and printable PDF voucher receipts in Indian Rupees (₹).
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2. Getting Started & Installation

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System Prerequisites

- **Node.js:** v18.0.0 or higher
- **Package Manager:** `npm` (v9.0.0+) or `yarn`
- **Database:** PostgreSQL v14+ (Optional; fallback in-memory mode active by default)

Installation & Launching the Application

1. Clone the Repository:

```
git clone https://github.com/Masthan008/edusyncai.git
cd "EduSync AI"
```

2. Backend Setup & Launch:

```
cd backend
npm install
npm run db:seed      # Seeds default initial data and demo credentials
npm run dev          # Launches server on http://localhost:5000
```

3. Frontend Setup & Launch:

```
cd ../frontend
npm install
npm run dev          # Launches UI client on http://localhost:3000
```

3. User Roles & Access Control

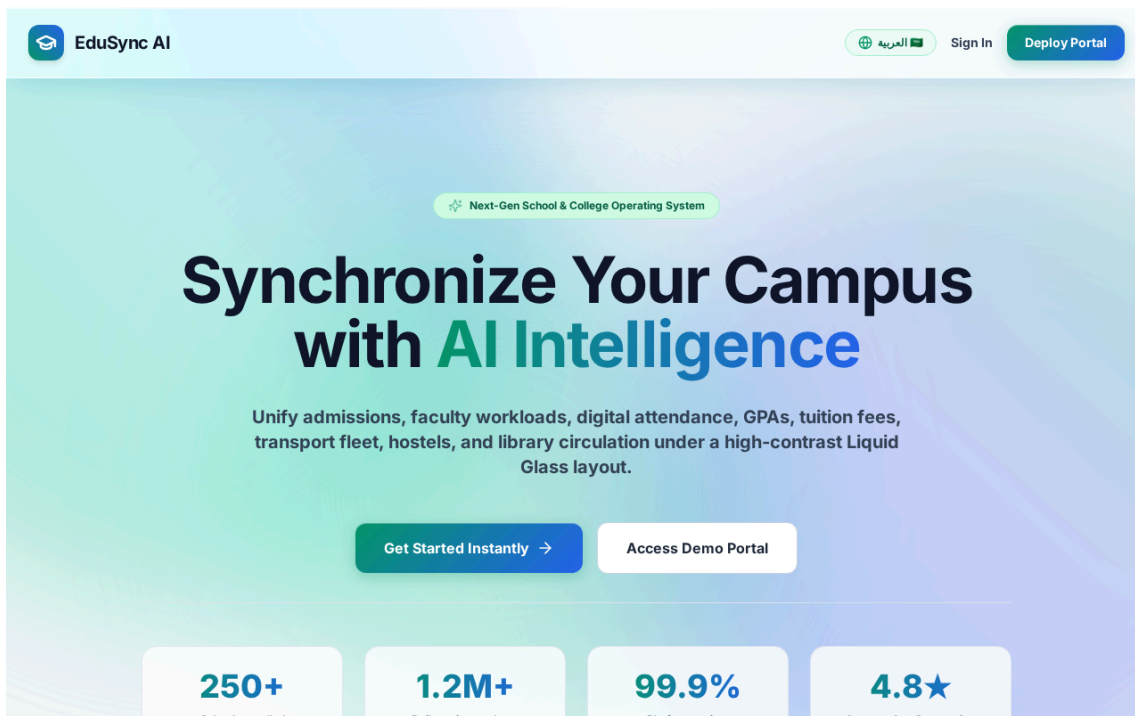
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EduSync AI implements Role-Based Access Control (RBAC) to ensure security and privacy across all institution operations.

Role	Access Level	Primary Responsibilities	Direct Section Link
Admin	System Wide	System configuration, database backups, pricing tier management, global user accounts	Admin Manual
Principal	Academic Executive	Institution oversight, SOP policy approval, staff/student analytics, curriculum monitoring	Principal Manual
HOD	Department Head	Departmental schedules, course allocations, teacher evaluations, subject performance	HOD Manual
Teacher	Classroom Level	Attendance check-ins, assignment creation, exam grading, student AI performance notes	Teacher Manual
Student	Personal Portal	Viewing timetables, submitting assignments, AI Tutor assistance, checking grades & attendance	Student Manual
Parent	Ward Guardian	Monitoring student progress, attendance alerts, fee invoice payment & receipt downloads	Parent Manual
Accountant	Financial Desk	Processing tuition fees, generating transaction ledger reports, managing pending balances	Accountant Manual

4. GUI Interfaces & Core Navigation

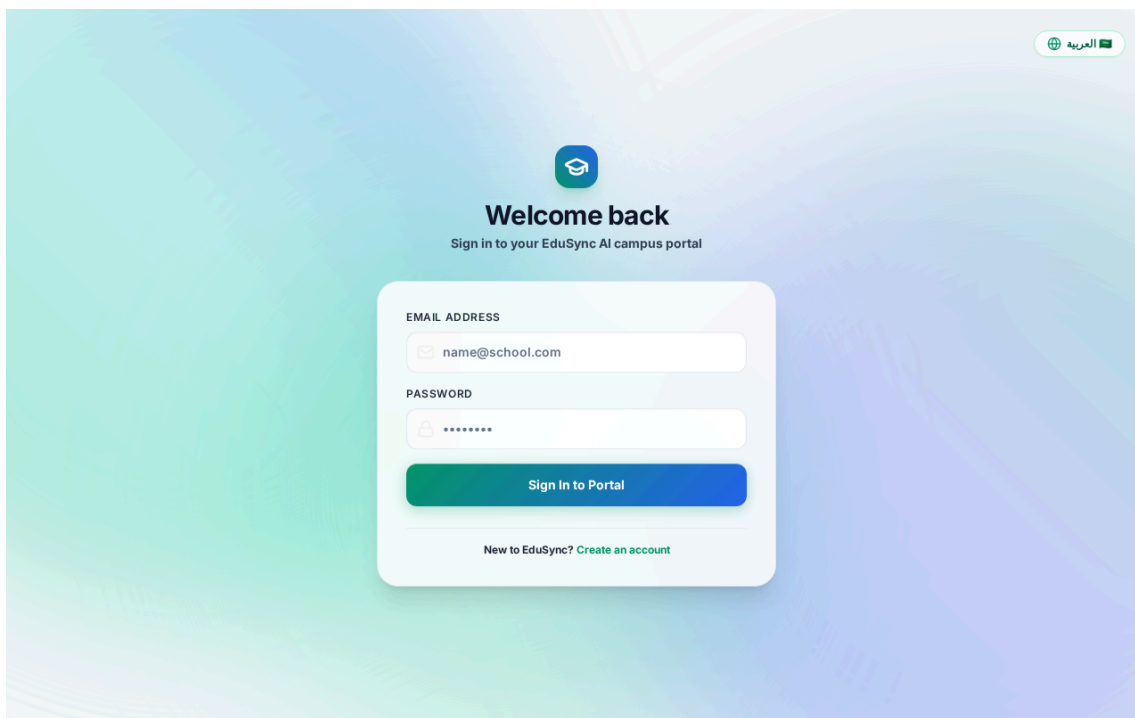
4.1 Institutional Public Landing Page



- **GUI Overview:** The public landing page features a modern Liquid Glass aesthetics with translucent frosted panels, live pricing tiers in Indian Rupees (₹), enterprise feature highlights, and direct sign-in entry points.
- **Key Navigation Points:**
 - **Header Bar:** Quick links to Pricing Tiers, PRD Documentation, and Language Switcher (English / العربية).
 - **Call to Action (CTA):** "Explore Portal" button redirects directly to the portal login screen.

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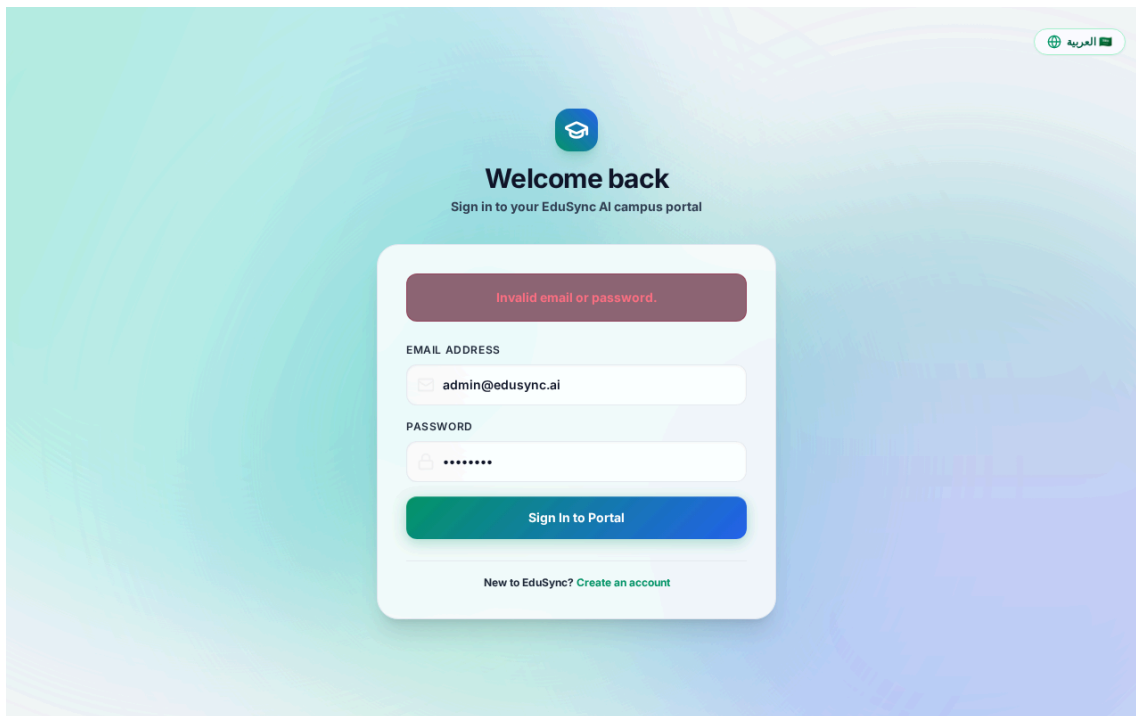
4.2 Unified Authentication Portal



- **GUI Overview:** Translucent glassmorphism login card centered on dynamic liquid blob animated backgrounds. Supports single sign-on authentication across all 7 user personas.
- **Default Credentials for Testing:**
 - **Admin:** admin@edusync.com / admin123
 - **Principal:** principal@edusync.com / admin123
 - **Teacher:** teacher@edusync.com / admin123
 - **Student:** student@edusync.com / admin123
 - **Parent:** parent@edusync.com / admin123
 - **Accountant:** accountant@edusync.com / admin123

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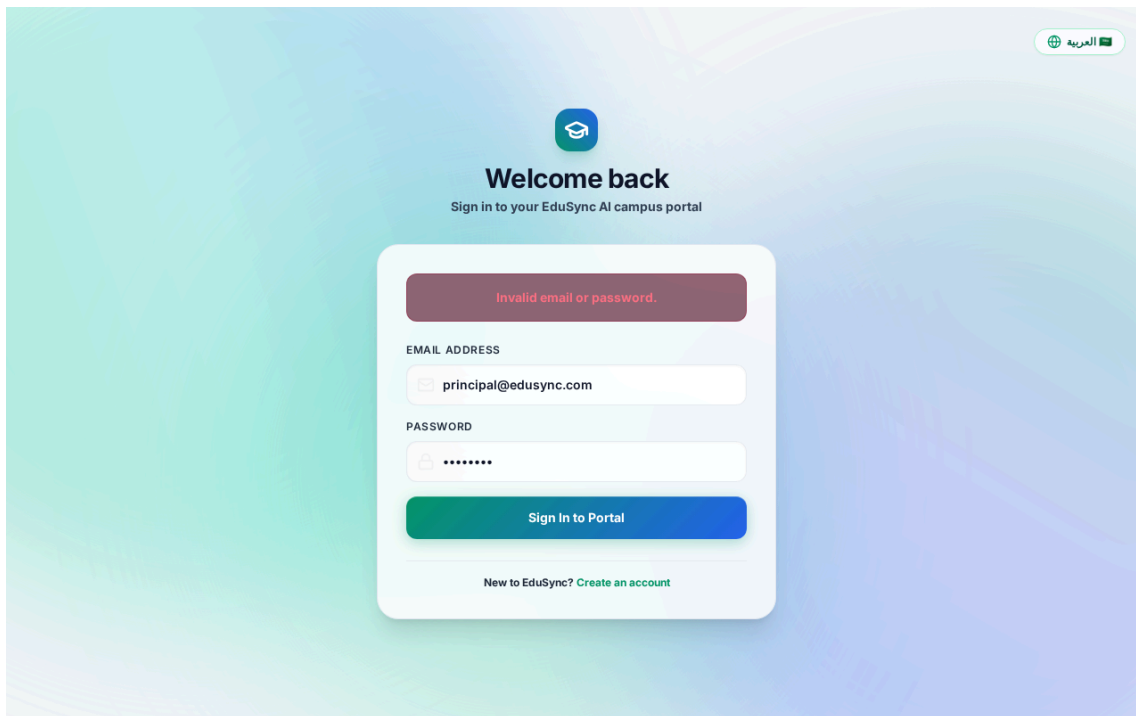
4.3 Administrator & Leadership Executive Dashboard



- **GUI Overview:** Real-time KPI summary dashboard displaying active student enrollment, total faculty count, pending tuition collections (₹), and system health metrics.
- **Key Navigation Points:**
 - **Sidebar Menu:** Access to [SIMS \(Students\)](#), [FIMS \(Teachers\)](#), [Attendance](#), [Grading](#), [Finance](#), [AI Tutor](#), and [SOP Guidelines](#).
 - **Bilingual Switcher:** Top-right language toggle converts interface instantly into Arabic Right-to-Left (RTL) layout.

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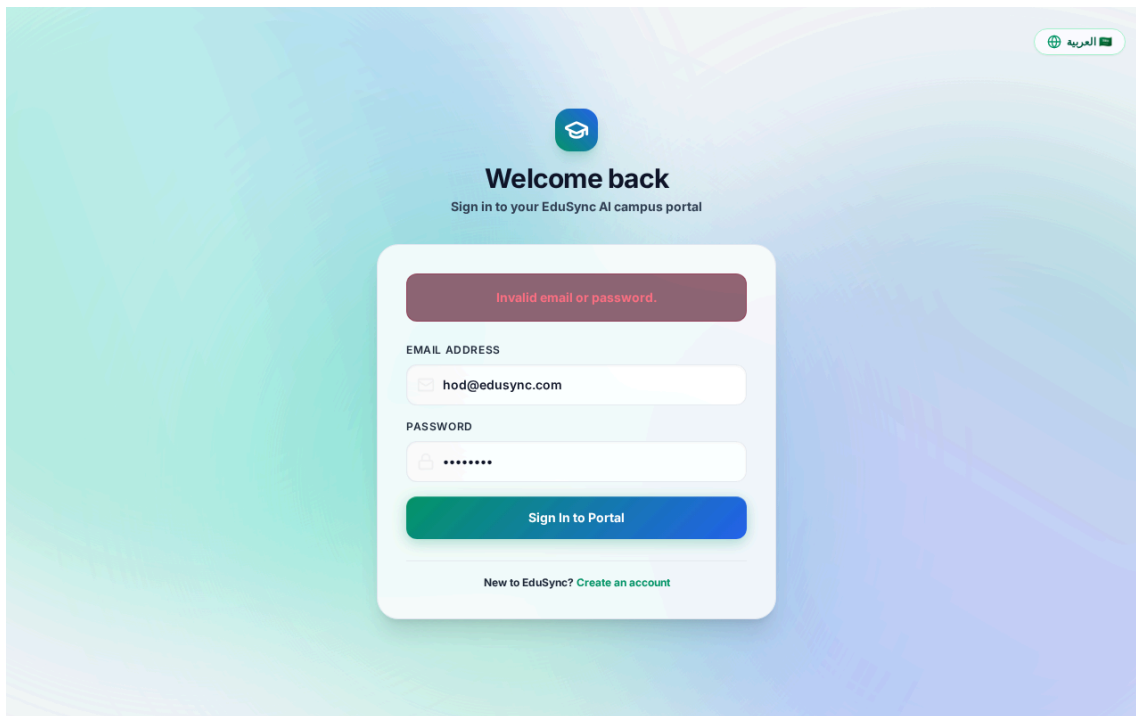
4.4 Principal & Leadership Portal



- **GUI Overview:** Provides academic institution oversight, performance heatmaps, teacher activity logs, and at-risk student monitoring.
- **Key Actions:** Approve institution SOPs, view multi-department pass rates, and trigger institutional reporting.

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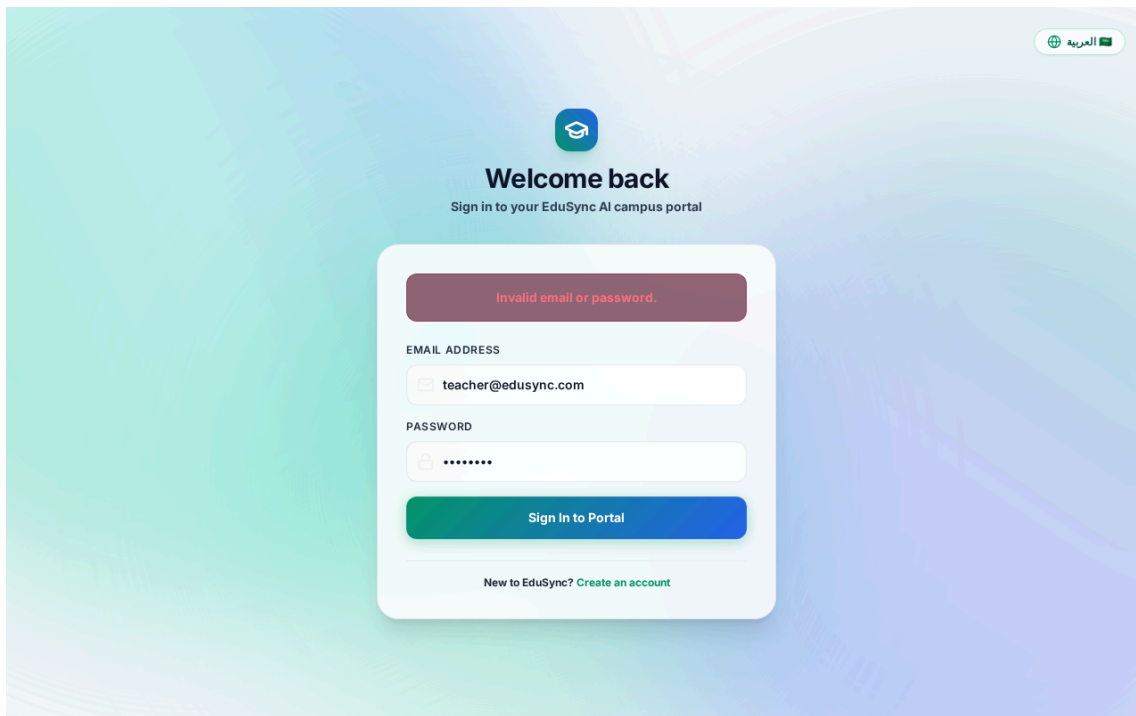
4.5 Department Head (HOD) Portal



- **GUI Overview:** Dedicated portal for managing course allocations, faculty timetables, and subject-wise grade distributions.
- **Key Actions:** Review subject exam results, assign period schedules, and review departmental guidelines.

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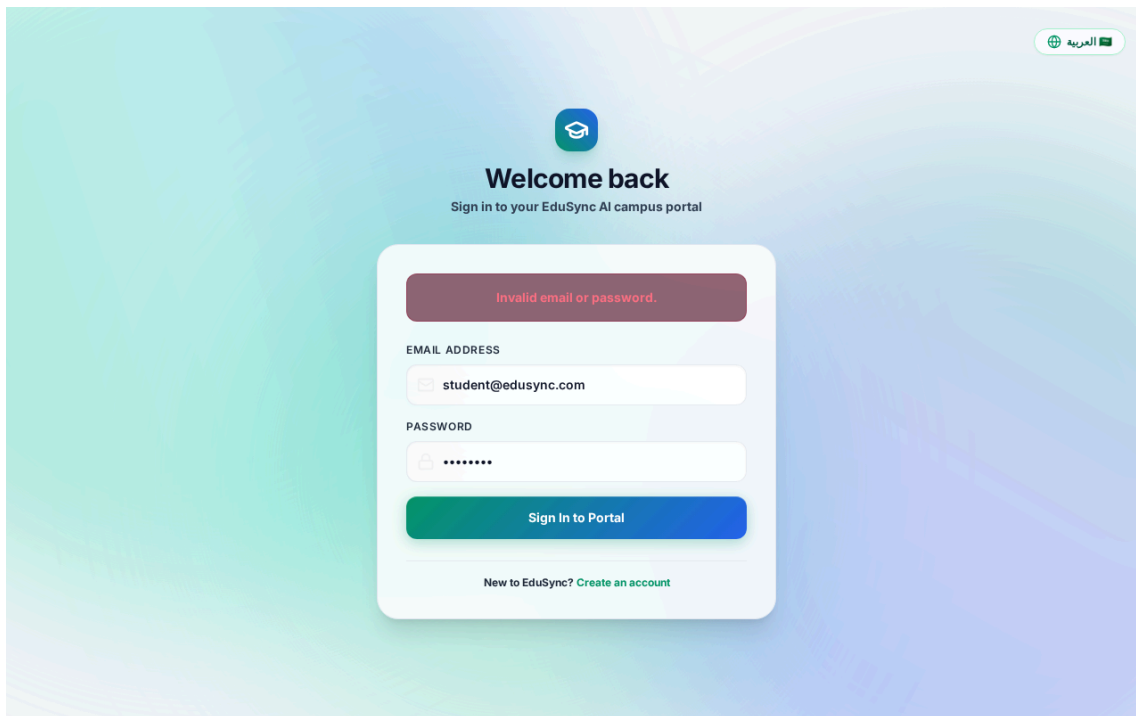
4.6 Teacher & Faculty Portal



- **GUI Overview:** Classroom-centric view allowing teachers to take digital attendance registers, publish homework assignments, and log test marks.
- **Key Actions:** Record daily attendance, input student exam scores, and check subject timetable grids.

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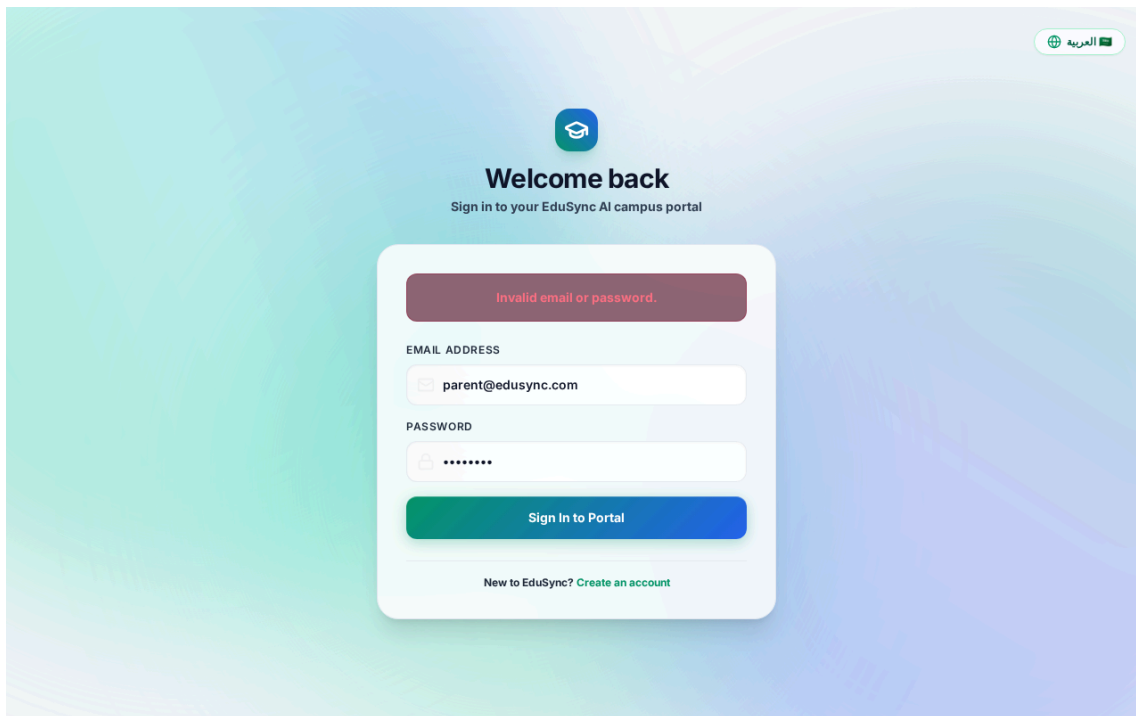
4.7 Student Portal & AI Tutor



- **GUI Overview:** Student workspace showing class schedules, attendance percentage, active GPA, assignment submission dropzones, and the Google Gemini 1.5 Flash AI Tutor chatbot.
- **Key Actions:** Chat with AI Tutor for homework help, submit assignment PDFs, and track transcript GPA.

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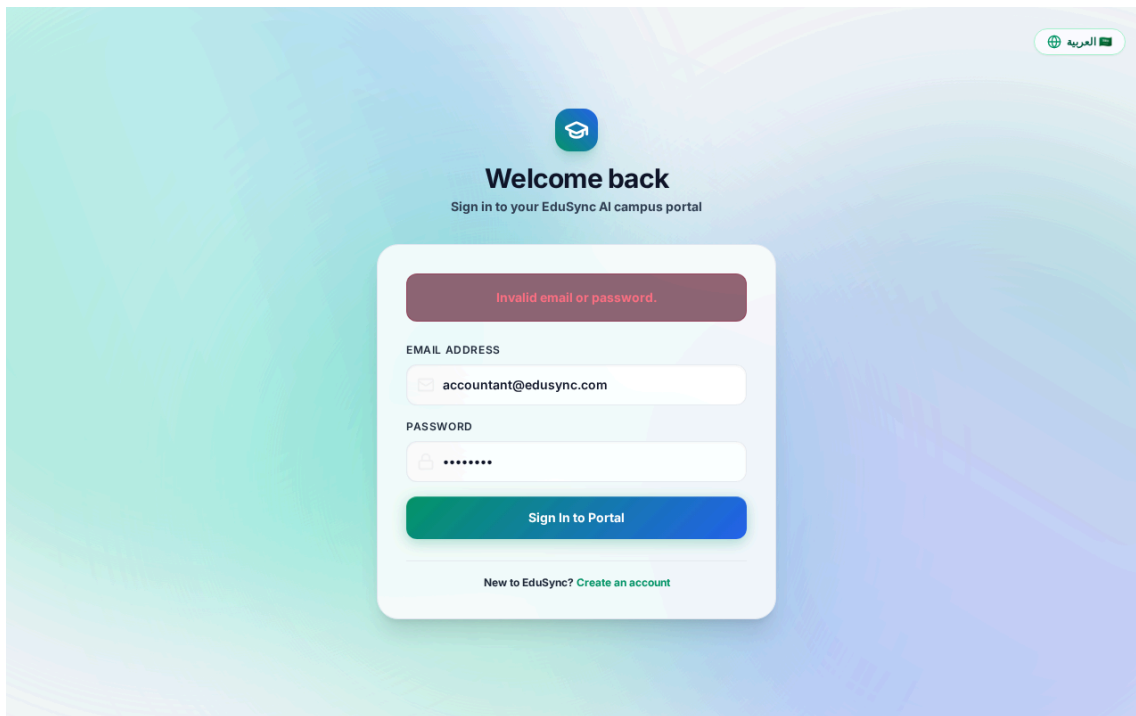
4.8 Parent & Guardian Portal



- **GUI Overview:** Transparent portal for guardians to monitor their ward's academic standing, attendance alerts, fee invoices, and download official payment receipts.
- **Key Actions:** Pay tuition fees in INR (₹), download PDF receipts, and communicate with class teachers.

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4.9 Accountant & Financial Desk



- **GUI Overview:** Financial management suite displaying total collected revenue, pending tuition dues, monthly collection graphs, and invoice generator tools.
- **Key Actions:** Issue student fee invoices, record manual payments, and export transaction ledgers.

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5. Comprehensive Module Operational Manual

Module 1: Student Information Management System (SIMS)

- **Overview:** Central database for student profiles, admission records, and parent linkages.
- **Operational Steps:**
 1. Go to **Students** section in the sidebar.
 2. Click **Admit Student**, fill in mandatory fields (Full Name, Roll Number, Class/Section, Guardian Details).
 3. Click **Save Student**. Profile becomes immediately active across attendance and grading rosters.

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Module 2: Faculty Information Management System (FIMS)

- **Overview:** Directory of teaching staff, qualifications, assigned departments, and course schedules.
- **Operational Steps:**
 1. Open **Faculty / Staff** menu.
 2. Click **Register Faculty**, assign Department (e.g. Science, Humanities) and subject expertise.

3. View active teaching workloads and assign HOD roles.

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Module 3: Academic Scheduling & Timetable Engine

- **Overview:** Conflict-free timetable engine for classes, teachers, and rooms.
- **Operational Steps:**
 1. Select **Timetables** from the HOD/Admin sidebar.
 2. Pick Class Section and Day of week.
 3. Assign Period Slot, Subject, Teacher, and Room. System blocks duplicate teacher/room assignments automatically.

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Module 4: Digital Attendance Register

- **Overview:** Daily and subject-wise attendance register with automated guardian alert triggers.
- **Operational Steps:**
 1. Teachers open **Attendance Register**.
 2. Select Class, Section, and Subject.
 3. Toggle student status (**Present, Absent, Late**) and click **Submit Register**.

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Module 5: Examinations, Grading & GPA Calculator

- **Overview:** Exam score logging with automated letter grade and 4.0 GPA scale conversion.
- **Operational Steps:**
 1. Open **Exams & Grading**.
 2. Select Term (e.g., Mid-Term 2025) and Subject.
 3. Enter student marks. System calculates percentage, Letter Grade, and GPA points automatically.

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Module 6: Assignments & Digital Submissions

- **Overview:** Digital homework distribution and student file upload portal.
- **Operational Steps:**
 1. Teacher creates assignment with title, description, and due date.
 2. Student opens assignment card, attaches document, and clicks **Turn In**.
 3. Teacher reviews submission, enters score, and posts feedback.

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Module 7: Finance, Tuition Fees & Invoicing (INR ₹)

- **Overview:** Multi-channel fee collection, invoice dispatch, and printable receipt vouchers in INR (₹).
- **Operational Steps:**
 1. Accountant generates itemized invoice in INR (₹).
 2. Parent logs into portal and executes fee payment.

3. Click **Download Receipt** to print formatted PDF receipt voucher.

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Module 8: EduSync AI Tutor & At-Risk Predictive Analytics

- **Overview:** Generative AI chatbot powered by Google Gemini 1.5 Flash and predictive student academic risk analytics.
- **Operational Steps:**
 1. Students open **EduSync AI Chat** for study aid, practice questions, and explanation.
 2. Principals view **At-Risk Analytics** tab to identify students requiring academic intervention.

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Module 9: SOP Policy Guidelines & Arabic RTL Support

- **Overview:** Institutional policy hub featuring 1-click AI translation into Arabic and native Right-to-Left (RTL) interface rendering.
- **Operational Steps:**
 1. Open **Policy & SOP Guidelines**.
 2. Click **Translate to Arabic (العربية)** for instantaneous Gemini AI translation.
 3. Click the top navbar language toggle to switch full UI to Arabic (`dir="rtl"`).

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Module 10: Transport Fleet & Bus Route Engine

- **Overview:** Vehicle records, bus driver contact details, pickup routes, and monthly transport fee billing.
- **Operational Steps:**
 1. Open **Transport Fleet**.
 2. Register vehicle details, define route stops, and assign monthly route fare in INR (₹).
 3. Link students to specific bus routes.

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Module 11: Hostel & Dormitory Accommodation System

- **Overview:** Dormitory room management, bed allocations, and room rent billing.
- **Operational Steps:**
 1. Open **Hostel Management**.
 2. Configure hostel blocks and room types (Single, Double, AC).
 3. Allocate students to vacant rooms and monitor occupancy.

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Module 12: Library Management System (LMS)

- **Overview:** Book cataloging, ISBN barcode tracking, issue/return transactions, and fine calculation.
- **Operational Steps:**
 1. Open **Library LMS**.

- 2. Add book title, author, and ISBN number.
- 3. Process book checkouts and returns against student roll numbers.

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6. Pricing Tiers & Subscription Management

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EduSync AI offers flexible tier subscription models catered to different institutional sizes with native Indian Rupee (INR ₹) pricing:

Plan Tier	Monthly Price	Target Institution	Included Capabilities
Academy Starter	₹9,999 / mo	K-12 Private Schools, Coaching Academies	Up to 500 Students, SIMS, FIMS, Attendance, Exams, Invoicing
Campus Enterprise	₹24,999 / mo	Colleges, High Schools, Multi-Branch	Unlimited Students, Gemini AI Tutor, At-Risk Analytics, Arabic RTL SOPs, Transport, Hostel, Library LMS
District SaaS Network	₹49,999 / mo	University Networks, Education Boards	Multi-tenant isolation, Custom Domain, SSO/SAML, On-Premises Docker Deployment, SLA Support

Administrators can update pricing tier details directly through the Admin Dashboard under `/api/school/pricing`.

7. Troubleshooting & Frequently Asked Questions (FAQ)

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Q1: What happens if the primary PostgreSQL database disconnects?

Answer: EduSync AI features an integrated **In-Memory Fallback Engine**. In the event of PostgreSQL database maintenance or connection drop, the system automatically switches to in-memory mode, ensuring continuous operational availability without crashing or showing error pages.

Q2: How does the AI translation for SOP guidelines work?

Answer: The SOP module sends the guideline document to the Google Gemini 1.5 Flash API endpoint. If external API access is restricted, local fallback translation engines automatically deliver standard Arabic translations.

Q3: How do parents download payment receipts?

Answer: Parents can log into the Parent Portal, navigate to **Fees & Payments**, select any invoice marked **PAID**, and click **Download Receipt**. A formatted PDF receipt with transaction ID and breakdown will be downloaded.

Q4: Can teachers change grades after publishing them?

Answer: Grades can be updated by the subject teacher or Department Head (HOD) prior to final term lock. Any modification made after lock requires approval from the Principal account.

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